BRAINHACK DONOSTIA 2026

RNN NIGHT FEVER

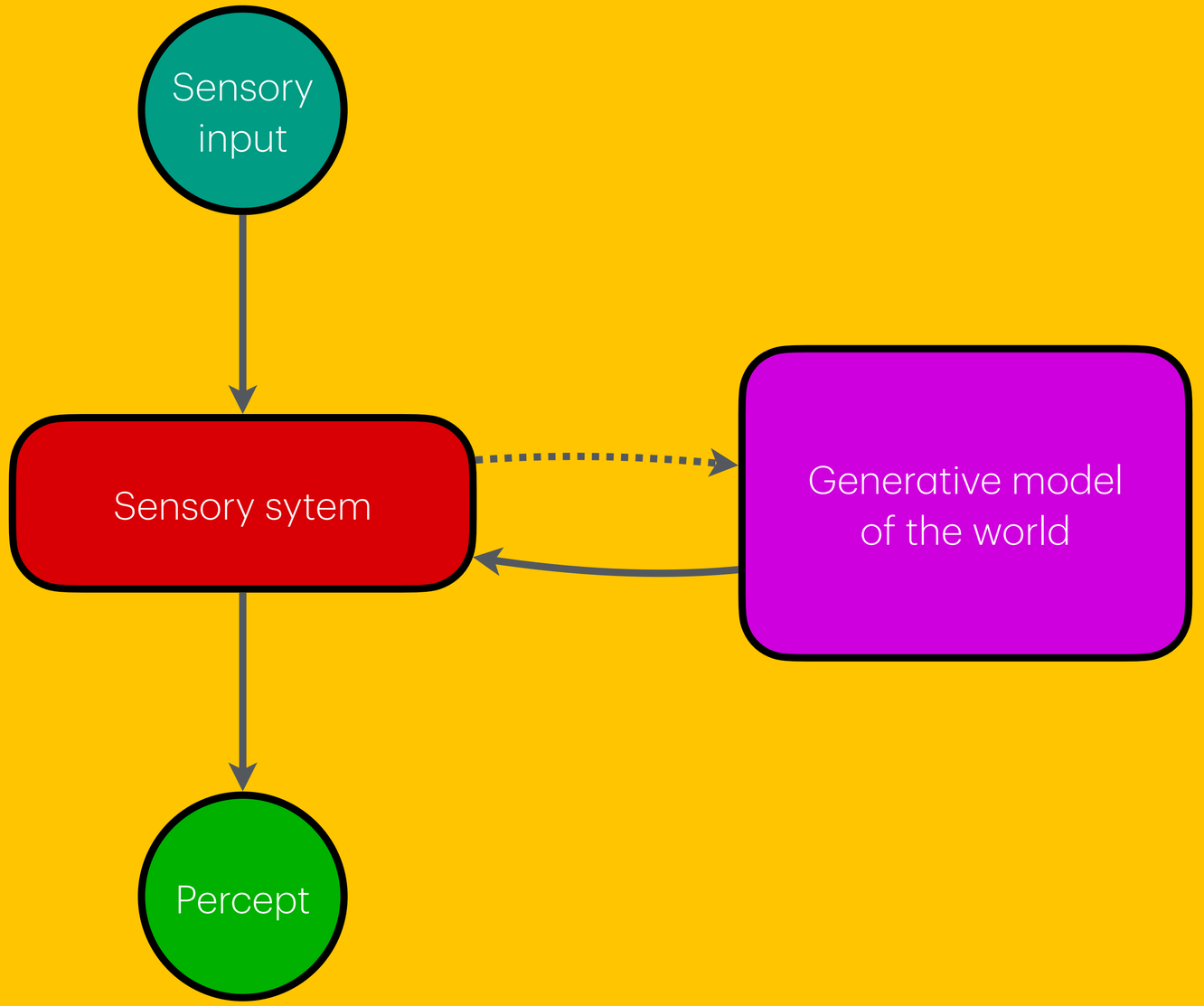
Does hippocampal replay emerge when recurrent neural networks go to sleep?

github.com/qtabs/rnn-night-fever



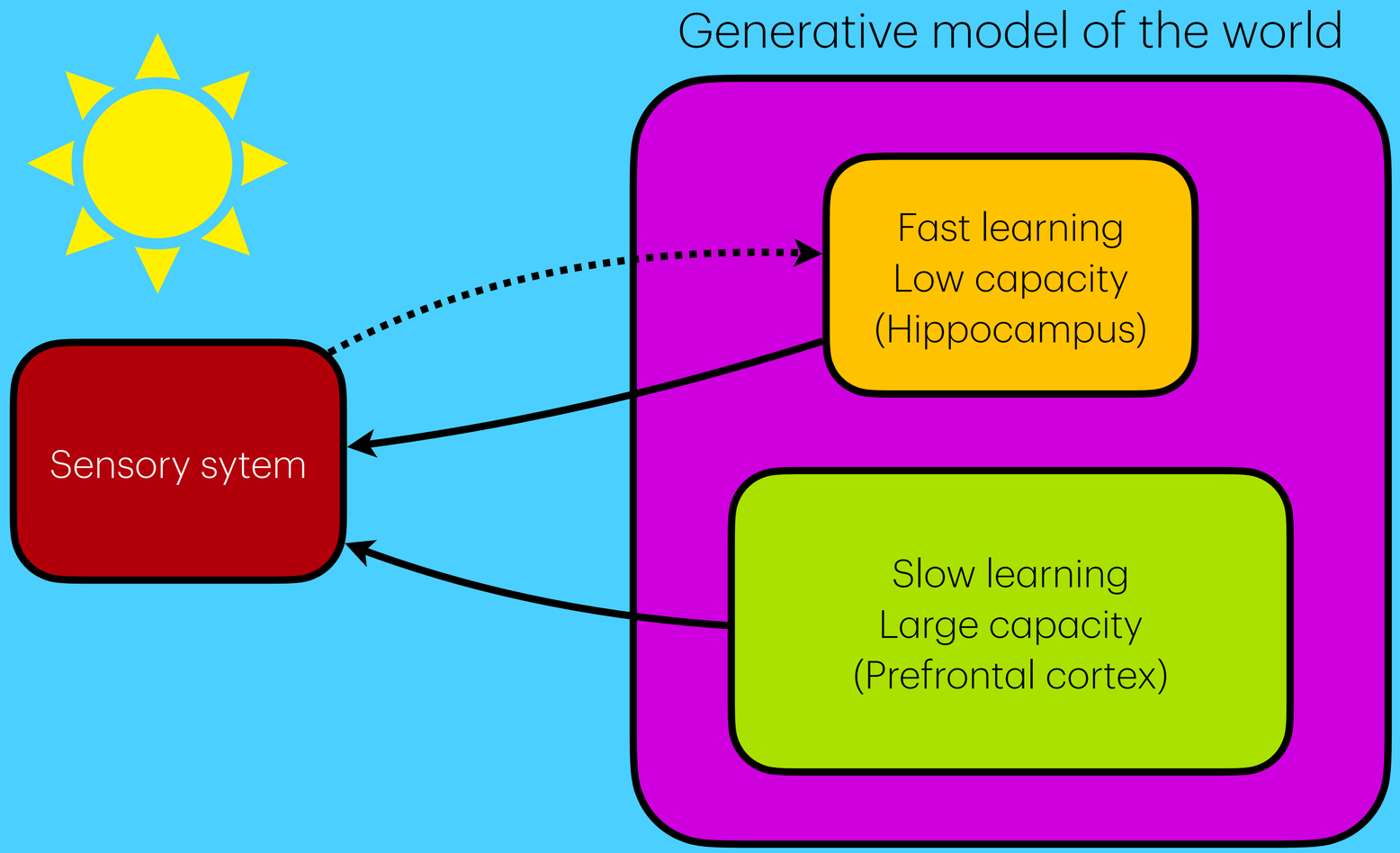
To perceive we first need to understand the world

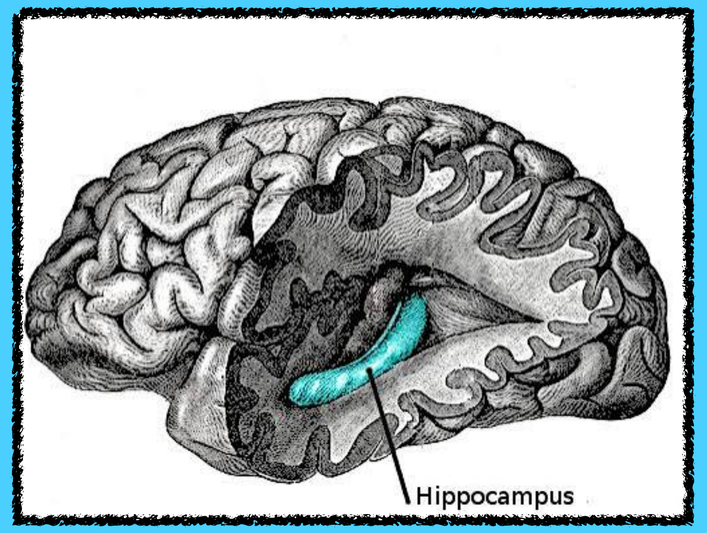




[Aitchison & Lengyel 2017 TICS]

But how do we learn the structure of the world?



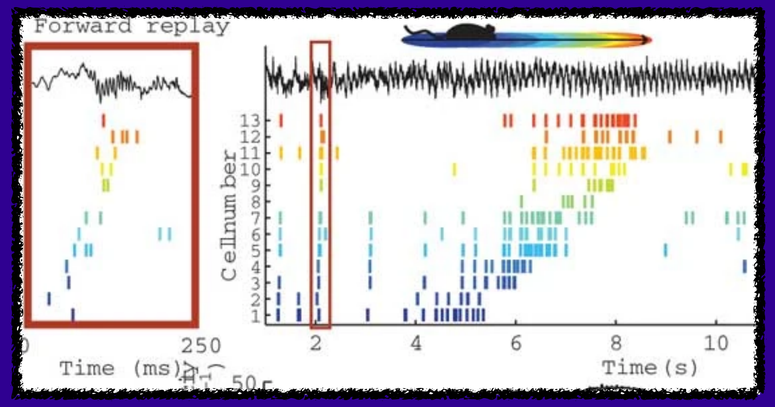


Henry Gray (1918) Anatomy of the Human Body, Plate 739

complementary systems hypothesis

[Singh, Norman, & Schapiro 2022 PNAS]

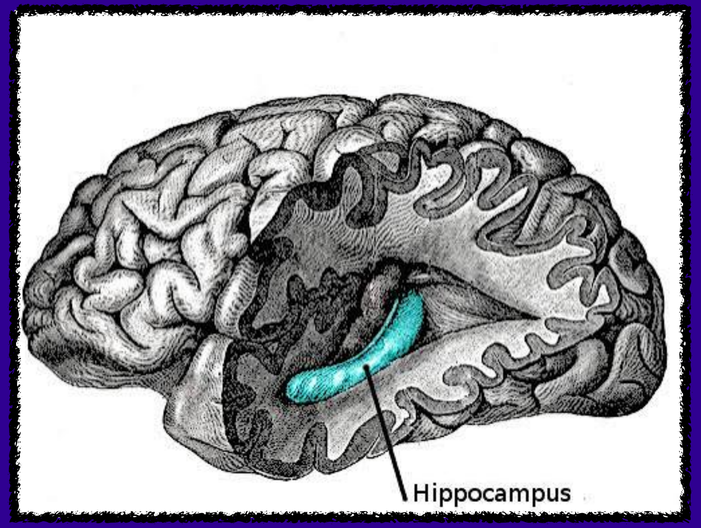
Memory consolidation through hippocampal replay



Carr, Jadhav & Frank; Nat Neuro 2011; Figure 1

Fast learning
Low capacity
(Hippocampus)

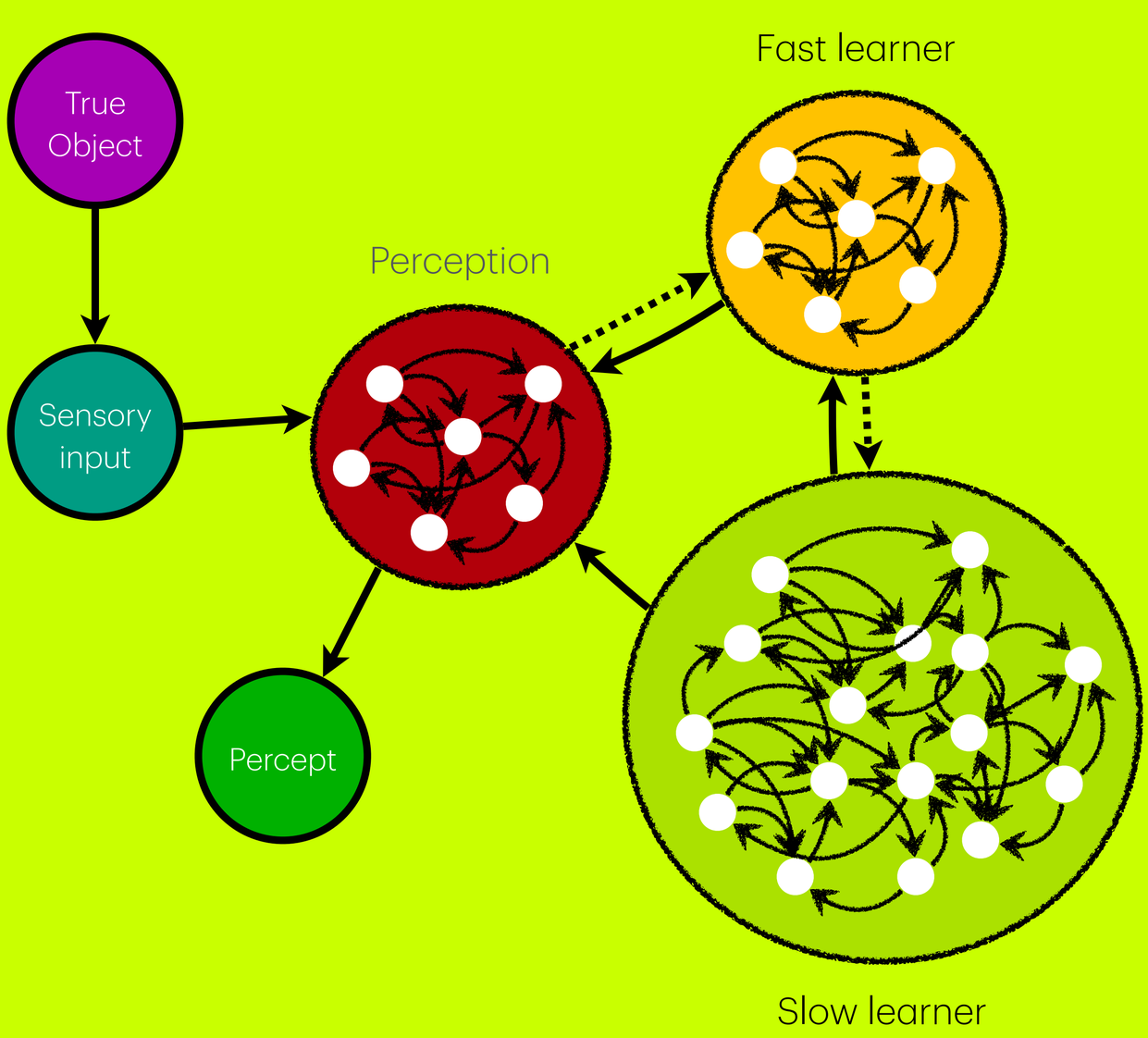
Slow learning
Large capacity
(Prefrontal cortex)

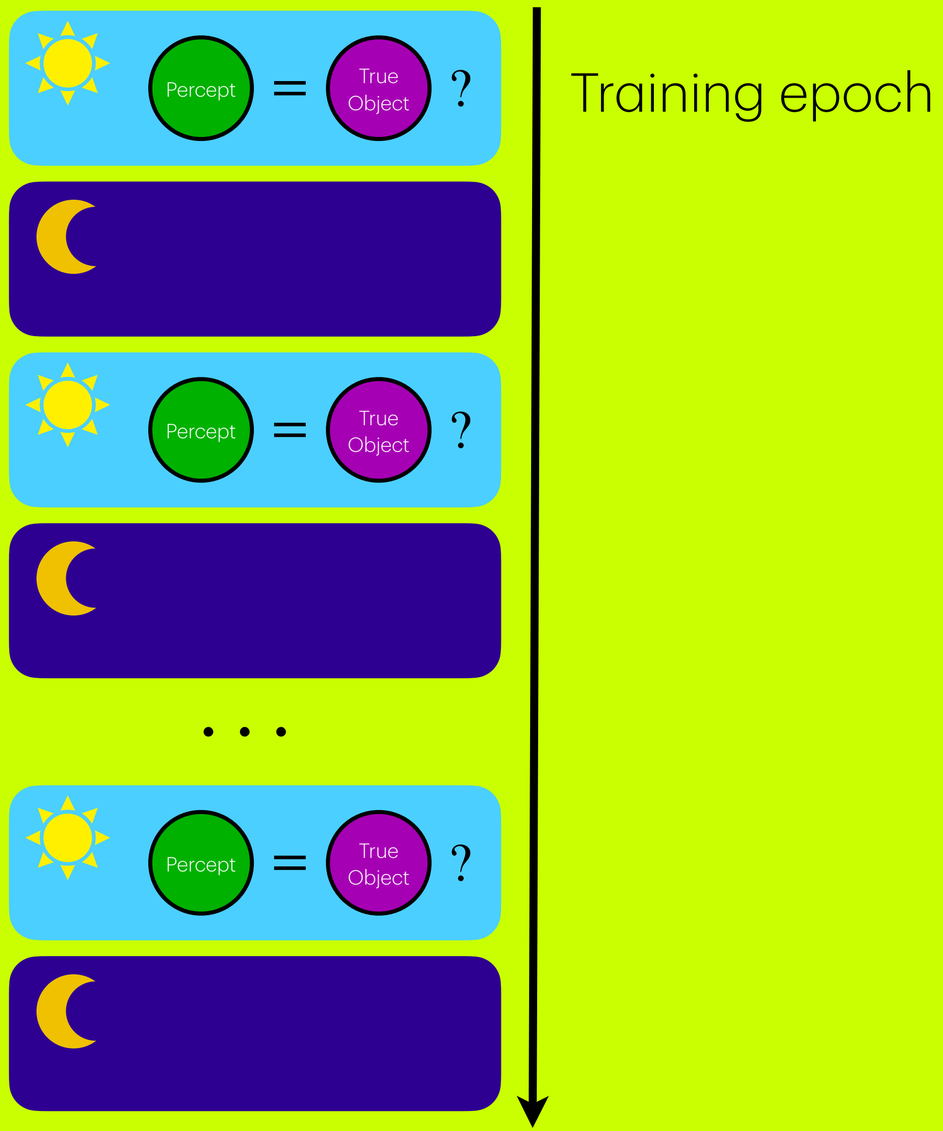


Henry Gray (1918) Anatomy of the Human Body, Plate 739

[Spens & Burgess 2024 HNB]

Is hippocampal replay the optimal learning strategy?

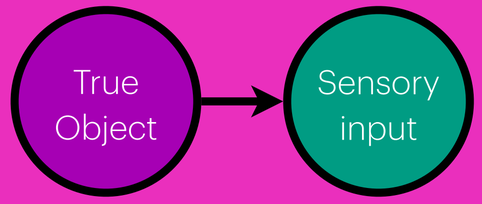




Three teams to keep the RNNs dancing all night long

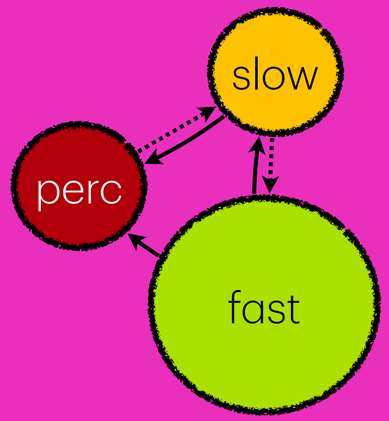
The Producers

- Design the structure of the ground-truth sensory world
- Generate sensory inputs
- Develop error functions to evaluate performance



The Architects

- Implement the architecture of the tree-network system
- Design the training strategy
- Develop the objective function and readouts



The Analysits

- Develop benchmark models and sufficiency criteria
- Implement analysis and visualisation scripts
- Write down the results



In conclusion:

- Hands-on neuroAI at its finest
- RNNs that sleep (and possibly snore)
- Bee-Gee themed presentations guaranteed
- CCN submission ready in 3 days!
- Absolutely unrealistic goals and timelines
- What else?

JOIN US!

github.com/qtabs/rnn-night-fever

